

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 - ELECTROSTATICS (3.4) - FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? Write your own Final Explanation of the mysterious balloon. Use evidence from your lessons: the balloon-and-hair data (20 strokes, hair rising about 5 cm, balloon sticking to the wall for about 3 minutes, paper pieces jumping at about 2 cm), the rods and strips, the triboelectric order, the conduction test, the electroscope readings, and the humidity data from Nairobi and Mombasa. Answer each section in full sentences. Use the words electric charge, electron transfer, electrostatic force, conductor, insulator, induction and earthing correctly. Draw labelled particle-level diagrams where asked. Say which lesson or observation each piece of evidence comes from. Remember that only electrons move in solids.

Part 1: Where do electric charges come from?

Describe the balloon-and-hair phenomenon in Lesson 1 in your own words. Then explain where the electric charges on the balloon and hair come from. In your answer: (a) describe what atoms contain and why a normal atom is neutral, using the charge values you met; (b) explain what happens to electrons and protons when the balloon is rubbed on dry hair; (c) use the triboelectric order to say why the balloon becomes negative and the hair positive; (d) explain why we say charge is conserved and not created, using the number of electrons moved and the number of atoms in a balloon.

Write your answer here:

Part 2: Why does the balloon attract or repel? Hair, balloon and the rule of forces

Explain why (a) the two rubbed balloons on threads pushed apart, and (b) the hair strands rise and are pulled toward the balloon. Use the results from the rods and strips (polythene strip and perspex rod) as evidence for the rule you use. Then explain why the balloon keeps its charge for about 3 minutes on a dry day rather than losing it at once, using what you found in the conduction test with the 3 V cell and bulb. Name the type of force involved.

Write your answer here:

Part 3: Why do neutral paper and the wall get attracted? Contact, induction and earthing

The paper pieces and the wall are neutral, yet the balloon attracts them. (a) Using a particle-level diagram and words, explain why neutral paper and wall are attracted but never repelled. (b) Explain why some paper pieces fall off after touching the balloon, using what you saw with the electroscope and the polythene strip touching the cap. (c) Compare charging by contact with charging by induction with earthing: which sign does each give, and what does earthing do? Use the electroscope observations as evidence.

Write your answer here:

Part 4: Charge on conductors, sparks, humidity and detecting charge

(a) Using the proof-plane and electroscope readings (hollow metal can; pear-shaped conductor), explain where charge sits on a conductor. Then explain how this accounts for the small shock felt on a dry Nairobi day (for example, Wanjiru's shock after combing her hair) and for lightning conductors in Kericho and Kisumu. (b) Explain why Amina's balloon hardly works in Mombasa but works well in dry Nairobi, using the humidity figures. (c) Explain how a gold-leaf electroscope shows that the balloon is charged, how you can tell the sign of a charge, and what a bigger leaf angle tells you.

Write your answer here:

Part 5: Applications and your complete Final Explanation

(a) Choose TWO applications from lightning conductors in Kericho and Kisumu, fuel tanker earthing at Nairobi petrol depots, photocopiers, and electrostatic precipitators at Athi River cement factories. For each one, explain which electrostatics idea from the unit it uses. (b) Now write a short, connected answer to the Driving Question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? Include hair, the wall, paper, humidity, and how your thinking changed from your Lesson 1 model.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Origin of charge (electron transfer and conservation)	Clearly explains that charges come from electrons already in atoms, that electrons move from hair to latex and protons stay put, uses the triboelectric order and the 1 nC, 6×10^9 electrons, 10^{23} atoms figures, and states that charge is transferred, not created.	Explains that electrons transfer from hair to balloon so the balloon is negative and hair positive, and mentions that charge is conserved, but gives few or no numbers or no reason from the triboelectric order.	Says that rubbing creates charge or does not say what moves, or mixes up which object is positive and which is negative.
Attraction and repulsion, including neutral objects	Uses like-repel and unlike-attract with evidence from the rods and strips, and explains attraction of neutral paper and wall by induction (charge separation, nearer charge attracts more), with a labelled particle diagram and the point that neutral objects are never repelled.	Applies the rule to balloons and hair and mentions that neutral objects are attracted because of charge separation, but with limited detail or no diagram.	Only describes attraction and repulsion without giving a cause, or says that neutral objects are repelled or that positive charge moves.

Conductors, insulators, charging methods and distribution	Correctly links free versus tightly bound electrons to the conduction test and to the balloon keeping its charge, and compares contact (same sign) with induction and earthing (opposite sign). Explains charge on outer surfaces and at points with point discharge, using the electroscope and proof-plane readings.	Correctly states the difference between conductors and insulators and between contact and induction, and gives some of the readings, with some gaps in explaining distribution or earthing.	Confuses conductors with insulators or contact with induction, or gives no use of the readings.
Detection, humidity and applications	Explains the electroscope (same sign on leaf and stem, bigger angle means more charge, sign test, earthing), explains humidity using 35% and 80% and the 3 minutes against under 30 seconds figures, and links two applications to the correct idea.	Explains how the electroscope detects charge and mentions humidity and one application correctly, with few data values.	Does not explain how the electroscope works or the effect of humidity, or applications are not linked to electrostatics ideas.
Use of evidence, vocabulary and coherence	Uses exact data and lesson evidence, uses terms such as electric charge, electrostatic force, discharge, earthing and induction accurately, and gives a connected final answer that shows how the Lesson 1 model has changed.	Uses most evidence and terms correctly and gives a mostly connected answer.	Uses little evidence or incorrect terms such as static electricity as an explanation, and the answer is disconnected.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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